Genus

Firmicutes/"Clostridia"/Clostridiales/Veillonellaceae/

Veillonella

Prévot 1933, 118[AL] emend. Mays, Holdeman, Moore, Rogosa and Johnson 1982, 35

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Veil.lo.nel'la. N.L. dim. suff. *-ella*; N.L. fem. n. *Veillonella* named after Adrien Veillon, the French microbiologist who isolated the type species.

Cocci, **0.3–0.5 μm in diameter**; cells are usually arranged in pairs, masses, or short chains. Stains **Gram negative**. Nonmotile, nonsporeforming. **Anaerobic**. Optimum temperature, 30–37°C. Optimum pH, 6.5–8.0. Oxidase negative. **Nitrate is reduced**. Some species produce an atypical catalase lacking porphyrin. **Gas is produced by most strains**. **Pyruvate**, **lactate**, malate, fumarate, and oxaloacetate are fermented. **Carbohydrates and polyols are not fermented by most strains**. Major metabolic end products in trypticase-glucose-yeast extract (TGY, see below) broth are **acetic and propionic acids**. In addition, **CO_2 and H_2** are produced from lactate (Rogosa, 1964). Resident of oral cavity, genito-urinary, respiratory, and intestinal tracts of humans and animals, but can also cause severe human infections such as bacteremia, endocarditis, osteomyelitis, and prosthetic joint infection.

DNA G+C content (mol%): 36–43 (T_m) or 40–44 (Bd).

Type species: **Veillonella parvula** (Veillon and Zuber 1898) Prévot 1933, 119[AL] (*Staphyloccocus parvulus* Veillon and Zuber 1898, 542).

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Number of validated species: 10

Further descriptive information

The genus *Veillonella* is represented by 10 species.

Morphology and ultrastructure

They are small, nonmotile, nonsporeforming organisms. Ultrastructure of thin sections viewed by electron microscopy reveal typical Gram-negative surface layers consisting of an outer membrane composed of two dense layers 3 nm wide separated by a less dense layer of 2 nm, a thin peptidoglycan layer, and a cytoplasmic membrane (Figure 1). By negative staining, the outer membrane appears extremely convoluted, whereas the peptidoglycan layer tightly follows the periphery of the protoplasm (Bladen and Mergenhagen, 1964).

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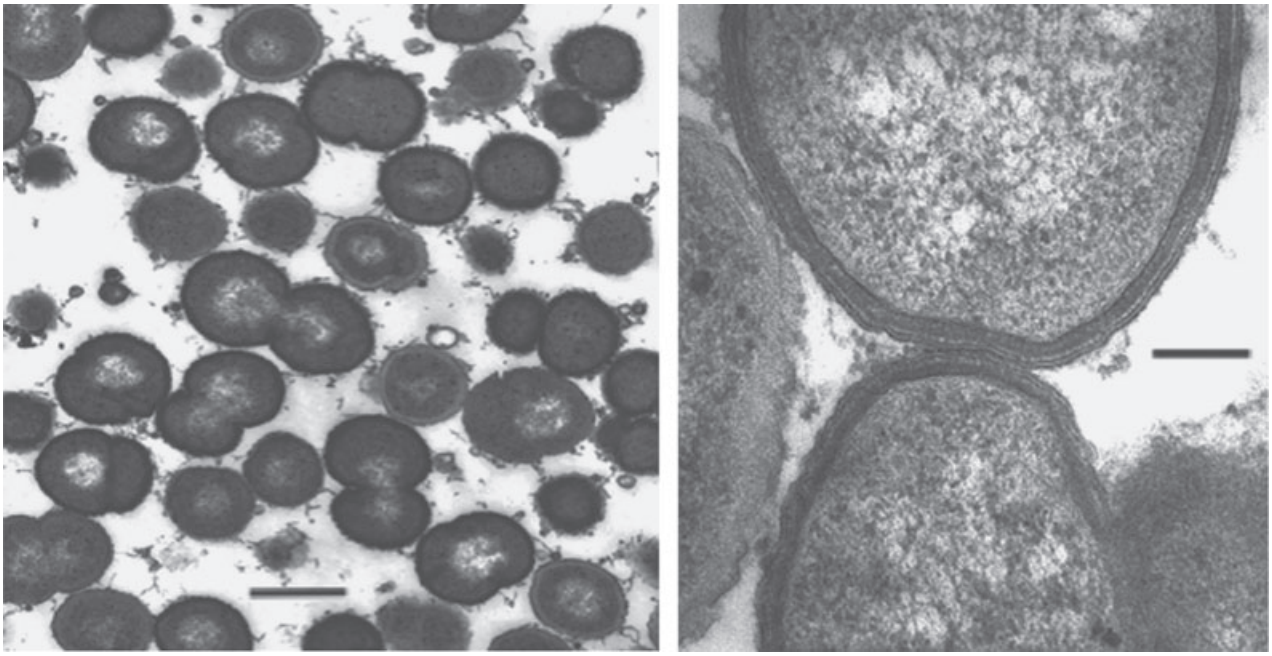
DOI: 10.1002/9781118960608.gbm00710. Published by John Wiley & Sons, Inc., in association with Bergey's Manual Trust.

FIGURE 1. Morphology and cell-wall structure of *Veillonella dispar* ATCC 17748. Transmission electron micrographs showing the general morphology after negative staining (left) and ultrathin section showing cell wall and membranes (right). Bars: left = 666 nm, right = 100 nm. (Courtesy of H. Marchandin.)



Colonial characteristics

Colonies on blood agar are 1–3 mm in diameter and appear smooth, opaque, and grayish-white. Nonhemolytic. *Veillonella*, unlike *Acidaminococcus*, *Anaeroglobus*, and *Megasphaera*, exhibits a pink to red fluorescence on brain heart infusion agar (BHI) containing either sheep or horse blood (Brazier and Riley, 1988). Fluorescence appears under UV light at 366 nm and fades rapidly after 5–10 min of exposure to air. However, the reaction depends upon the medium and varies among the species. The fluorescent pigment is a porphyrin.

Nutrition and metabolism

The species of *Veillonella* are not clearly distinguished by their phenotypic characteristics. Most species, except *Veillonella criceti* and an unidentified phylogenetically related strain (Marchandin et al., 2005), are unable to ferment carbohydrates or amino acids (Rogosa, 1964). Some strains require putrescine or cadaverine (Rogosa and Bishop, 1964a). H_2S is produced from cysteine, cystine, glutathione, thiosulfate, thiocyanate, and thioglycolate (Rogosa and Bishop, 1964b). Indole is not produced; gelatin is not liquefied. Lactate enhances the growth with accompanying production of acetic and propionic acids, CO_2, and H_2. Rogosa and Bishop (1964b) reported that succinate is metabolized by resting cells, whereas the organisms cannot grow with succinate as an energy source in basal medium. However, most strains of *Veillonella* species ferment succinate naturally present within TGY broth. In addition, when succinate is added to trypticase-yeast extract broth, it is metabolized to propionate. In fact, succinate is co-metabolized with lactate, and the decarboxylation energy is mainly used for nongrowth purposes (Samuelov et al., 1990).

About 34% of strains are catalase positive, and 99% of strains are susceptible to kanamycin and colistin special-potency antibiotic disks (Jousimies-Somer et al., 1999). Resistance to colistin has been observed for *Veillonella montpellierensis*, the type strain of *Veillonella ratti* (Jumas-Bilak et al., 2004), and an unidentified phylogenetically related strain (Marchandin et al., 2005).

Major metabolic end products from TGY broth are acetic and propionic acids, which are produced in approximately equal quantities.

Fatty acid composition

Major cellular fatty acids are $C^{13:0}$ and $C^{17:1}$ ω^8. Minor cellular fatty acids include $C^{11:0}$, $C^{12:0}$, $C^{14:0}$, $C^{15:0}$, $C^{16:1}$ ω9c, $C^{16:0}$, $C^{17:0}$, $C^{18:1}$ ω9c, $C^{18:0}$. Other compounds are present in trace amounts (Table 1).

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TABLE 1. Relative content (%) of cellular fatty acids of *Veillonella* species[a]

Fatty acid	*V. parvula* ATCC 17745^T	*V. atypica* ATCC 17744^T	*V. caviae* DSM 20738^T	*V. criceti* DSM 20734^T	*V. denticariosi* RBV 106^T	*V. dispar* ATCC 17748^T	*V. montpellierensis* ADV 281.99^T	*V. ratti* ATCC 17746^T	*V. rodentium* ATCC 17743^T
11:00	1.8	1.1	1.4	2	0.5	0.8	2.3	2.1	1.6
12:00	4.9	3.8	4	1.4	3	3.8	7	4.5	5
13:00	24	20.4	7.3	16.3	12.9	23.5	6.3	20	25.3
14:0 iso	1.9	0.6	1.2	–	5.1	1.5	–	0	–
14:00	6.5	5.7	3.1	1.2	3.9	4.9	5	2	9.2
15:00	12	17.7	7	12.7	4.4	7	6.3	10.3	11.6
14:0 2OH	–	0.3	–	0.3	–	–	–	1	–
16:1 ω9c	5.2	5.2	10.6	4.2	4.5	7.7	8.4	4.8	5.5
16:00	7	7.5	10.3	3.7	8.6	8.4	12.9	4.6	6.7
17:1 ω8	22.3	25.7	29.9	48.9	10.1	25.6	20.7	36.9	17
17:00	4.1	2.4	2.5	2.6	3.2	2.5	2	2.4	4.2
18:1 ω9*c*	6.2	4.3	14.3	2.8	5.7	8	17.4	6.1	6.7
18:1 ω9*t*	–	–	1.9	1.2	3.5	–	2	–	–
18:00	3.9	2.7	6.2	2.1	6.6	5.7	11.1	5.8	6.7

[a]Fatty acid nomenclature: unsaturated fatty acids, the position of the double bond can be located by counting from the methyl (ω) end of the carbon chain; OH, hydroxylated fatty acids; *cis*, *trans* isomers are indicated by the suffix *c* and *t*, respectively.

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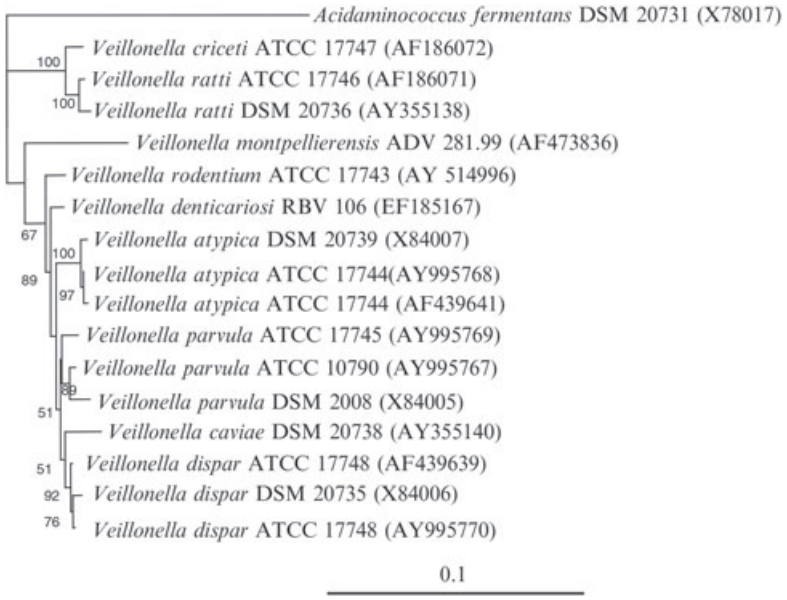
FIGURE 2. Phylogeny of *Veillonella* species. The phylogenetic relationships of all sequences of type strains of the genus *Veillonella* available from public databases are shown above. The sequence of *Acidaminococcus fermentans* DSM 20731 was used as an outgroup. Bar = 10% difference in 16S rRNA gene sequences. The neighbor-joining method was used for tree construction. One hundred bootstrap trees were generated, and bootstrap confidence levels are shown as percentages on the nodes.



Genetics

Veillonella species can be differentiated by DNA–DNA hybridization (Mays et al., 1982), 16S rRNA gene sequence comparison (Figure 2), and restricted fragment-length polymorphism (RFLP) analysis of 16S rRNA genes (Sato et al., 1997). However, the utility of the last two methods is limited by the low level of sequence variation among some *Veillonella* species and the frequent intrachromosomal heterogeneity between copies of the 16S rRNA genes in human isolates (Marchandin et al., 2003c). Four 16S ribosomal operons are present in reference strains of *Veillonella* parvula, *Veillonella dispar*, *Veillonella atypica*, and 27 clinical isolates. The variability observed between these operons is higher than that between the 16S rRNA gene sequences of Veillonella parvula and *Veillonella dispar*. Thus, 16S rDNA-based methods may be not suitable for the identification of these species. A similar situation occurs within the type strains of *Veillonella ratti* and *Veillonella criceti* which form a very tight phylogenetic group. Thus, analyses of 16S rRNA gene sequences can not identify species within this group (Marchandin et al., 2005). The limitations of 16S rRNA gene sequence analyses within *Veillonella* and related genera caused by the variability of the 16S rRNA operons have been reviewed in detail (Marchandin and Jumas-Bilak, 2006). The 70 kDa heat-shock protein gene (*dnaK*) is suggested to be an effective alternative for discriminating between species because of its higher interspecies variability.

The G + C values for the DNA are 36–43 mol%; the majority clustering around 39 mol% when tested by the thermal denaturation method (Mays et al., 1982).

Genomic structure

The genome of *Veillonella* species consists of one circular chromosome of 2.15 Mb (Marchandin et al., 2003a).

Plasmids and bacteriophages

Plasmids have been found in about 50% of the human oral strains. Their molecular sizes range from 1.1 to 28 MDa (Arai et al., 1984). Mays et al. (1982) have suggested that the fructose fermentation by some strains of *Veillonella criceti* may be associated with the presence of a plasmid.

A temperate bacteriophage specific for genus *Veillonella* has been reported (Shimizu, 1968). In addition, 25 virulent phages were isolated from washings of the oral cavity of 200 individuals (Hiroki et al., 1976; Totsuka, 1976). These phages were classified into two major groups on the basis of their morphology and serological characteristics. Seven phages infected strain ATCC 17743, now classified as *Veillonella*

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rodentium. The lipopolysaccharide of *Veillonella rodentium* ATCC 17743 acts as receptor for one of the bacteriophages (Totsuka, 1988; Totsuka and Ono, 1989).

Antibiotic susceptibility

Veillonella species are considered susceptible to ampicillin (0.12–4 μg/ml), amoxycillin/clavulanic acid (0.06–2 μg/ml), piperacillin/tazobactam (0.12–32 μg/ml), cefoxitin (0.12–8 μg/ml), cefotetan (0.12–2 μg/ml), ceftriaxone (0.12–8 μg/ml), imipenem (0.06–1 μg/ml), meropenem (0.06–0.12 μg/ml), clindamycin (0.06–0.12 μg/ml), bacitracin (1–4 units/ml), and metronidazole (0.5–4 μg/ml) and are resistant to ramoplanin (512 μg/ml), trimethoprim/sulfamethoxazole (8–256 μg/ml), and vancomycin (32–256 μg/ml) (Finegold et al., 2004; Roberts et al., 2006).

In the first edition of *Bergey's Manual of Systematic Bacteriology* (Rogosa, 1984b), veillonellae are considered susceptible to penicillin G. However, based on the National Committee for Clinical Laboratory Standards (NCCLS, now called Clinical and Laboratory Standards Institute) recommendations (National Committee for Clinical Laboratory Standards, 2003), most strains should be considered resistant. Several studies have shown that a large population of *Veillonella* species is penicillin resistant with MIC over 2 μg/ml (the breakpoint for susceptibility according to NCCLS criteria). Resistance to amoxycillin and penicillin has also been reported for *Veillonella dispar* and *Veillonella parvula* (Ready et al., 2004). In another study on 40 clinical strains tested by agar dilution method, 80% were found resistant to penicillin (Reig et al., 1997). The strains for which penicillin MICs were >8 μg/ml were also less susceptible to ampicillin and cefoxitin (MIC >0.25μg/ml and >1 μg/ml, respectively). The mechanism involved in high levels of penicillin resistance in *Veillonella* remains to be elucidated. Although occasional strains of *Veillonella parvula* produce β-lactamase (Valdes et al., 1982), the main resistance mechanism of veillonellae may be due to alterations in a penicillin-binding protein (Theron et al., 2003).

Antibiotic resistance genes

Tetracycline resistance (Tcr) genes are present within about 10–12.5% of *Veillonella* species isolated from the oral cavity (Lancaster et al., 2005, 2003; Ready et al., 2006). Five different Tc^r genes have been detected: *tetA*, *tetL*, *tetM*, *tetO*, and *tetS*. The most common is *tetM*, followed by *tetS*. Some strains harbor two Tc^r genes (Ready et al., 2006). The *tetM* gene is located on a Tn*916*-like element.

A nitroimidazole resistance gene (*nimE*) is present in a metronidazole-susceptible *Veillonella* species (Marchandin et al., 2004). This gene is probably located on a low copy number plasmid which can explain the susceptibility observed.

Pathogenicity

Veillonellae are commensals in the oropharynx, gastrointestinal tract, and uro-genital tract of humans and animals. Of the ten recognized species, only *Veillonella parvula*, *Veillonella dispar*, *Veillonella atypica*, *Veillonella denticariosi*, and *Veillonella montpellierensis* have been isolated from humans. However, a strain genetically closely related to the *Veillonella criceti*–*Veillonella ratti* group has been isolated from a human clinical specimen (Marchandin et al., 2005). Veillonellae are implicated in several systemic diseases such as meningitis (Bhatti and Frank, 2000), bone infections (Isner-Horobeti et al., 2006; Singh and Yu, 1992), prosthetic joint infections (Marchandin et al., 2001; Zaninetti-Schaerer et al., 2004), endocarditis (Boo et al., 2005; Houston et al., 1997; Rovery et al., 2005), bacteremia (Liu et al., 1998), and in various infections in children (Brook, 1996).

Pathogenicity in animals is unknown.

Virulence factors

Mergenhagen et al. (1961) established that a phenol-water extract of *Veillonella* cells exhibited biological and immunological activities characteristic of endotoxins and that this water-soluble material was composed mainly of lipid and polysaccharide. The lipid moiety accounting for approximately two-thirds of the complex contains glucosamine, tridecanoic acid, 3-hydroxytridecanoic acid and 3-hydroxypentadecanoic acid. (Hewett et al., 1971; Tortorello and Delwiche, 1984). Tridecanoic acid and 3-hydroxytridecanoic acid represent about 50% of the total lipopolysaccharide fatty acids (Bishop et al., 1971). The polysaccharide moiety contains 2-keto-3-deoxyoctonic acid (KDO), glyceromannoheptose, galactose, rhamnose, glucose, and glucosamine. (Hewett et al., 1971; Hofstad and Kristoffersen, 1970). However, other hexoamines may be present (Hewett et al., 1971). The *Veillonella* LPS is highly endotoxic. Sveen showed that *Veillonella* LPS induces skin inflammation and the Schwartzman phenomenon in rabbits (Sveen, 1977). Moreover, phenol-water extracted lipopolysaccharide is lethal at nanogram levels for mice, chick embryos, pyrogenic in rabbits, and gelated *Limulus* amoebocyte lysate. The endotoxin activity of *Veillonella* LPS is comparable to that from *Fusobacterium* and *Salmonella enteritidis* (Sveen et al.,

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1977). LPS from both *Veillonella parvula* and *Veillonella atypica* reduce microbial activity of human neutrophils but do not affect monocyte function (Focà et al., 1990).

Ecology

Veillonellae colonize the human mouth within the three first months of life (Kononen et al., 1992). The colonization is not dependent on the presence of erupted teeth (Kononen et al., 1999). The organisms persist in the oral flora, and *Veillonella* species remain the dominant anaerobic bacteria in the saliva of elderly edentulous persons (Sato et al., 1993). Dental plaque, buccal mucosa, and the tongue are the main ecological niches of *Veillonella* species.

During the plaque development, the dominant strains of *Veillonella* change in their phenotypic and genotypic characteristics (Palmer et al., 2006). Within the oral cavity, *Veillonella* species form coaggregates with many other bacteria (Hughes et al., 1988). Coaggregation promotes formation of mixed-species bacterial colonies between veillonellae and other oral bacteria. Both the rapid succession within the *Veillonella* population and coaggregation are important processes in establishment of a stable oral ecosystem (Hughes et al., 1988; Palmer et al., 2006).

Enrichment and isolation procedures

Veillonellae can be isolated by streaking samples onto Columbia or Wilkins-Chalgren sheep-blood agar and incubating at 37°C for 2 d. Lactate agar medium containing vancomycin (7.5 μg/ml) favors isolation. Strict anaerobic conditions are required for growth.

Maintenance procedures

Strains may be maintained in trypticase-glucose-yeast extract (TGY) medium (Carlier et al., 2002) under anaerobic conditions as described above. Cultures of veillonellae will usually survive for no longer than 1 week without subcultivation. Strains in the exponential growth stage in BHI or TGY broth can be stored at −80°C or in liquid nitrogen. Cells suspended in skim milk can be successfully lyophilized.

Differentiation of the genus *Veillonella* from closely related taxa

The differentiation of the genus *Veillonella* from other genera of anaerobic Gram-negative cocci can be done primarily on the basis of poor growth with gas formation on the usual media (except for *Veillonella denticariosi* which does not produce gas), nitrate reduction, and lactate fermentation. Other characteristics are given in Table 1 of the chapter describing the genus *Anaeroglobus*.

Upon routine laboratory examination, *Veillonella* is frequently misidentified as *Dialister*, a tiny Gram-negative rod. Nevertheless, a positive nitrate reaction and production of large amounts of propionic acid are the key features differentiating *Veillonella* species from most *Dialister* species which are negative for these tests. Although *Dialister propionicifaciens* produces small amounts of propionic acid, it does not reduce nitrate to nitrite. Upon misinterpretation of the Gram reaction, *Parvimonas micra*, which is a very small anaerobic Gram-positive coccus, can also be misidentified as *Veillonella*. *Parvimonas micra* does not produce propionic acid and does not reduce nitrate.

Taxonomic comments

The type species of the genus *Veillonella* was first isolated by Veillon and Zuber in 1898 and designated *Staphylococcus parvulus*. Prévot (1933) further described these bacteria and proposed the genus *Veillonella*, with two species, *Veillonella parvula* and *Veillonella alcalescens*. He placed the new genus in the family *Neisseriaceae*. Based on serological reactions, Rogosa (1965) divided veillonellae into seven distinct subspecies, which were subsequently shown to be species by DNA–DNA hybridization (Mays et al., 1982; Rogosa, 1984b). In 1971, the genus *Veillonella* was excluded from the family *Nesseriaceae* and transferred into the new family *Veillonellaceae* (Rogosa, 1971b). Subsequently, it was proposed to include the genus *Veillonella* in a new family "*Acidaminococcaceae*" (Garrity and Holt, 2001). The family "*Acidaminococcaceae*" was created to include members of the *Firmicutes* or low G+C Gram-positive bacteria that possessed a Gram-negative type of cell wall. *Acidaminococcus* was selected as the type genus. However, the placement of the genus *Veillonella* (Prévot, 1933) in the family "*Acidaminococcaceae*" is not in accordance with the Rules 23a and 56a of the International Code of Nomenclature of Bacteria (1990 Revision) (Lapage et al., 1992). Moreover, the name *Veillonellaceae* has priority over "*Acidaminococcaceae*". Therefore, in the current volume, the family "*Acidaminococcaceae*" is not used, and the family *Veillonellaceae* is retained.

Today, the identification of *Veillonella* at the species level remains difficult. Because of the lack of phenotypic criteria, the commonly used biochemical tests are useless. Serological tests (Rogosa, 1965) are no longer available. DNA–DNA hybridization (Mays et al., 1982), which seems to be a good method, is laborious, difficult to interpret and therefore

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not well-suited for routine identification. Molecular genetic methods, such as RFLP analysis and 16S rRNA sequencing, are valuable means to distinguish some species (Kolenbrander, 2006). However, these methods, though helpful, are not absolute. There is doubt as to whether *Veillonella parvula* and *Veillonella dispar* (Marchandin et al., 2003c; Palmer et al., 2006)as well as *Veillonella criceti* and *Veillonella ratti* (Marchandin et al., 2005) are truly separate species. Alternative molecular markers to 16S rRNA, such as other housekeeping genes, could clarify these interspecies relationships. For instance, three highly conserved genes, *rpoB*, *dnaK*, and *gyrB*, had greater discriminatory power than the 16S rRNA gene in differentiating between *Veillonella* strains. Of these, the *rpoB* gene showed the highest level of interspecies but the lowest intraspecies diversity (Byun et al., 2006). Thus, multilocus Sequencing Typing (MLST) of highly conserved genes could be an attractive method to differentiate *Veillonella* species (Stackebrandt et al., 2002).

List of species of the genus *Veillonella*

Veillonella parvula (Veillon and Zuber 1898) Prévot 1933, 119[AL] (*Staphyloccocus parvulus* Veillon and Zuber 1898, 542)

par'vu.la. L. fem. dim. adj. *parvula* very small.

The characteristics are as described for the genus. Some strains of *Veillonella parvula* can grow aerobically in static culture on lactate (Kolenbrander, 2006). The type strain of *Veillonella parvula* subsp. *parvula* ATCC 10790 has been found to have 74% homology with type strain of *Veillonella alcalescens* subsp. *alcalescens* ATCC 17745 and, therefore, they were considered as synonyms (Mays et al., 1982). *Veillonella parvula* has priority.

Antigenic properties: serogroup VI (*Veillonella parvula* subsp. *parvula*); serogroup IV (*Veillonella alcalescens* subsp. *alcalescens*) (Rogosa, 1965).

Pathogenicity: *Veillonella parvula* has been associated with osteomyelitis, prosthetic valve endocarditis, and bacteremia. Two fatal cases of *Veillonella parvula* bacteremia have been reported (Liu et al., 1998). Isolated from oral cavity and intestinal tract of humans and rodents.

DNA G+C content (*mol%*): 37–40, mean 38 (T_m) and 41 (Bd).

Type strain: ATCC 10790, ATCC 17742, CCUG 5123, DSM 2008, JCM 12972, KCTC 5019, NCTC 11810, VPI 11221, Prévot Te 3, ATCC 17745, VPI 11224.

GenBank accession number (*16S rRNA gene*): AY995767, X84005 (*Veillonella parvula* subsp. *parvula*); AY995769 (*Veillonella parvula* subsp. *alcalescens*).

Veillonella atypica (Rogosa 1965) Mays, Holdeman, Moore, Rogosa and Johnson 1982, 35[VP] (*Veillonella parvula* subsp. *atypica* Rogosa 1965, 707)

a.ty'pi.ca. Gr. pref. *a* not; L. adj. *typicus -a -um* typical; N.L. fem. adj. *atypica* not typical.

The characteristics are as described for the genus.

Antigenic properties: Serogroups V and VI (Rogosa, 1965).

Pathogenicity: not reported; isolated from human saliva.

DNA G+C content (*mol%*): 36–40, mean 39 (T_m).

Type strain: ATCC 17744, DSM 20739, NCTC 11830, VPI 11220 (strain KON).

GenBank accession number (*16S rRNA gene*): AF439641, AY995768, X84007.

Veillonella caviae Mays, Holdeman, Moore, Rogosa and Johnson 1982, 35[VP]

ca.vi'a.e. N.L. gen. n. *caviae* of *Cavia*, the zoological genus name of the guinea pig.

The characteristics are as described for the genus.

Antigenic properties: Serogroup VIII (Rogosa, 1984b).

Pathogenicity: not reported; isolated from guinea pig mouth.

DNA G+C content (*mol%*): 37–39, mean 39 (T_m).

Type strain: ATCC 33540, DSM 20738, NCTC 12021, VPI 12140, Rogosa strain PV1.

GenBank accession number (*16S rRNA gene*): AY355140.

Veillonella criceti (Rogosa 1965) Mays, Holdeman, Moore, Rogosa and Johnson 1982, 35[VP] (*Veillonella alcalescens* subsp. *criceti* Rogosa 1965, 708)

cri.ce'ti. N.L. gen. n. *criceti* of *Cricetus*, the zoological genus name of the hamster.

The characteristics are as described for the genus. Nine out of ten strains fermented fructose (Rogosa, 1984b). A human clinical isolate phylogenetically closely related to *Veillonella criceti* and fermenting fructose has been reported (Marchandin et al., 2005).

Antigenic properties: Serogroup I (Rogosa, 1965).

Pathogenicity: not reported; isolated from hamster mouth.

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DNA G+C content (*mol%*): 38–40, mean 39 (T_m).

Type strain: ATCC 17747, DSM 20734, VPI 11226, NCTC 12020, Rogosa strain HV1.

GenBank accession number (*16S rRNA gene*): AF186072.

Veillonella denticariosi
Byun, Carlier, Jacques, Marchandin and Hunter 2007, 2847[VP]

den.ti.car.i.o'si. L. n. *dens dentis* tooth; L. *adj. cariosus* rotten, decayed; N.L. gen. n. *denticariosi* of a decayed tooth.

Does not produce gas from TGY broth. Other characteristics are as described for the genus. Can be differentiated from other species of the genus *Veillonella* by 16S DNA and *dnaK* sequencing.

Pathogenicity: Isolated from human carious dentin.

DNA G+C content (*mol%*): unknown.

Type strain: RBV 106, CIP 109448, CCUG 54362, DSM 19009.

GenBank accession number (*16S rRNA gene*): EF185167.

Veillonella dispar
(Rogosa 1965) Mays, Holdeman, Moore, Rogosa and Johnson 1982, 35[VP] (*Veillonella alcalescens* subsp. *dispar* Rogosa 1965, 708)

dis'par. L. fem. adj. *dispar* dissimilar, different.

The characteristics are as described for the genus.

Antigenic properties: Serogroup VII (Rogosa, 1965).

Pathogenicity: *Veillonella dispar* has been associated with prosthetic valve endocarditis and prosthetic joint infections. Isolated from mouth and respiratory tract of humans.

DNA G+C content (*mol%*): 38–40, mean 39 (T_m) and 42 (Bd).

Type strain: ATCC 17748, DSM 20735, NCTC 11831, VPI 11223, Rogosa strain ERN.

GenBank accession number (*16S rRNA gene*): AF439639, AY995770, X84006.

Veillonella montpellierensis
Jumas-Bilak, Carlier, Jean-Pierre, Teyssier, Gay, Campos and Marchandin 2004, 1315[VP]

mont.pel.li.er.en'sis. N.L. fem. adj. *montpellierensis* pertaining to Montpellier, where the type strain and two other strains were isolated.

The characteristics are as described for the genus. Cells are coccoid (0.3–0.5 mm in diameter) and occur singly, in pairs, or in short chains. Colonies on Columbia blood agar are 1–3 mm in diameter and appear smooth, opaque, and grayish-white. Can be differentiated from other species of the genus *Veillonella* by 16S DNA and *dnaK* gene sequencing.

Pathogenicity: *Veillonella montpellierensis* has been isolated from the gastric fluid of a newborn and from the amniotic fluid of 2 women. Association with endocarditis has been reported (Rovery et al., 2005).

DNA G+C content (*mol%*): unknown.

Type strain: ADV 281.99, CCUG 48299, CIP 107992, DSM 17217.

GenBank accession number (*16S rRNA gene*): AF473836.

Veillonella ratti
(Rogosa 1965) Mays, Holdeman, Moore, Rogosa and Johnson 1982, 35[VP] (*Veillonella alcalescens* subsp. *ratti* Rogosa 1965, 708)

rat'ti. N.L. gen. n. *ratti* of *Rattus*, the zoological genus name of the rat.

The characteristics are as described for the genus.

Antigenic properties: Serogroup III (Rogosa, 1965).

Pathogenicity: not reported; isolated from rat mouth.

DNA G+C content (*mol%*): 41–43, mean 42 (T_m) and 44 (Bd).

Type strain: ATCC 17746, DSM 20736, VPI 11225, NCTC 12019, Rogosa strain RV-12X.

GenBank accession number (*16S rRNA gene*): AF186071, AY355138.

Veillonella rogosae
Arif, Do, Byun, Sheehy, Clarke, Gilbert and Beighton 2007, 584[VP]

rog.o.sae. N.L. gen. n. *rogosae* of Rogosa, named in honor of the American microbiologist Morrison Rogosa.

The characteristics are as described for the genus. Using the rapid ID 32A system, alkaline phosphatase and pyroglutamic acid arylamidase activities are detected. Can be differentiated from other species of the genus *Veillonella* by 16S rRNA and *rpoB* sequencing.

Pathogenicity: not reported. Found in supra-gingival plaque of caries-free children.

DNA G+C content (*mol%*): unknown.

Type strain: 100CF, CCUG 54233.

GenBank accession number (*16S rRNA gene*): EF108443.

Veillonella rodentium
(Rogosa 1965) Mays, Holdeman, Moore, Rogosa and Johnson 1982, 35[VP] (*Veillonella parvula* subsp. *rodentium* Rogosa 1965, 707)

ro.den'ti.um. N.L. pl. gen. n. *rodentium* (from L. part. adj. *rodens -entis* gnawing), of rodents.

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The characteristics are as described for the genus.

Antigenic properties: Serogroup II (Rogosa, 1965).

Pathogenicity: not reported; isolated from hamster mouth.

DNA G+C content (*mol%*): 42–43, mean 43 (T_m) and 44.4 (Bd).

Type strain: ATCC 17743, DSM 20737, VPI 11222, NCTC 12018, Rogosa strain HV 19.

GenBank accession number (*16S rRNA gene*): AY514996.

Acknowledgements

I would like to thank Hélène Marchandin for her critical reading of this chapter and providing the transmission electron micrographs and Jean Euzéby for his advice on etymology.

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